3. Scenario: You are working on a project that involves analyzing a dataset containing information about houses in a neighborhood. The dataset is stored in a CSV file, and you have imported it into a NumPy array named house_data. Each row of the array represents a house, and the columns contain various features such as the number of bedrooms, square footage, and sale price.

Question: Using NumPy arrays and operations, how would you find the average sale price of houses with more than four bedrooms in the neighborhood?

AIM:

To calculate the average sale price of houses with more than four bedrooms using NumPy arrays.

PROCEDURE:

1. Import the NumPy library.
2. Create a NumPy array containing the number of bedrooms, square footage, and sale price of each house.
3. Select houses having more than four bedrooms using a condition.
4. Calculate their average sale price using np.mean().
5. Display the average sale price.

PROGRAM CODE:

```
import numpy as np
house_data = np.array([
    [3, 1200, 200000],
    [5, 1800, 350000],
    [4, 1500, 250000],
    [6, 2200, 450000],
    [5, 2000, 400000]
])
selected_houses = house_data[house_data[:, 0] > 4]
average_price = np.mean(selected_houses[:, 2])
print("Average Sale Price:", average_price)
```

INPUT:

```
[[3, 1200, 200000],
 [5, 1800, 350000],
 [4, 1500, 250000],
 [6, 2200, 450000],
 [5, 2000, 400000]]
```

OUTPUT:

```
import numpy as np

house_data = np.array([
    [3, 1200, 200000],
    [5, 1800, 350000],
    [4, 1500, 250000],
    [6, 2200, 450000],
    [5, 2000, 400000]
])

selected_houses = house_data[house_data[:, 0] > 4]

average_price = np.mean(selected_houses[:, 2])

print("Average Sale Price:", average_price)
```

```
Average Sale Price: 400000.0
```

RESULT:

Thus, houses with more than four bedrooms were selected using NumPy array operations. The average sale price of these houses is 400,000.